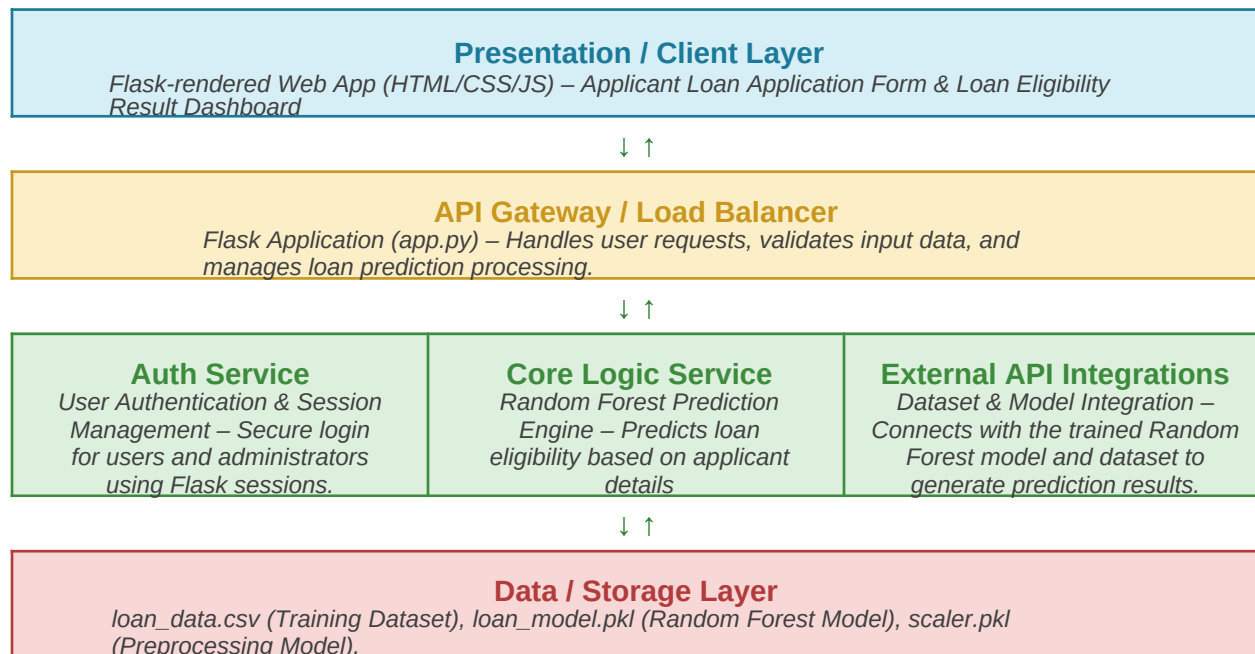


Solution Architecture

Date	30 June 2026
Team ID	*****
Project Name	Smart Lender – AI-Powered Loan Eligibility Prediction System
Maximum Marks	5 Marks

Solution Architecture Diagram:

The architecture shows the flow of applicant data through the AI-Powered Loan Eligibility Prediction System. It explains how the web application, machine learning model, and data storage work together to predict loan eligibility.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects applicant details through a web interface and displays the loan eligibility prediction result	HTML5, CSS3, JavaScript, Flask Templates
API Gateway	Handles HTTP requests, validates user input, invokes the prediction model, and returns the output.	Python, Flask

Component Name	Description / Role in Architecture	Technologies Used
Machine Learning Module	Processes applicant data using the trained Random Forest model and predicts loan eligibility.	Scikit-learn, Joblib, NumPy, Pandas
Data Storage	Stores the loan dataset used for training and the serialized machine learning model for prediction	CSV Dataset, Pickle (.pkl) File